

Seminar 1 – System Maintenance and Testing of IT systems

System Maintenance

Before reading the literature:

To me, system maintenance is about managing and supporting the IT systems that organizations use to perform various tasks. These are systems that the company has specifically requested and that are designed to fulfill certain functions. It's the responsibility of an IT team to ensure these systems continue to work reliably and serve their intended purpose without failure. Since technology is constantly evolving due to various factors, these systems need to be regularly maintained and updated to keep up with new requirements and changes.

After reading the literature:

Now I understand that system maintenance is much broader and more complex. Riggs (1969) shows that it is about keeping systems aligned with user and regulatory requirements. Edwards et al. (1984) expands the scope, showing that maintenance involves adapting and improving all elements of a system, not just repairing software. Finally, Nordström and Welanders (2005) connect maintenance directly to business value, stressing that maintenance must both preserve stability and enable change. My view has therefore shifted from a narrow technical perspective to seeing system maintenance as a strategic, continuous process that supports both IT operations and business development.

Quote 1 – Edwards (1984)

"Systems maintenance is defined here as an alteration to any system's element necessary to eliminate a fault in that element, adapt to a change in that or any other element, or to improve upon the performance characteristics of an element." (Edwards et al., 1984, s. 241).

Motivation:

This definition is broader than Riggs' and is valuable because it highlights that system maintenance isn't limited to software or hardware. Edwards frames it as an integrated activity that can involve any part of the system, from users and data to procedures and security. It shows how maintenance is about adaptation and improvement, not just fixing errors.

Quote 2 – Nordström & Welanders (2005)

"The work of continually managing, supporting, preserving and further developing permanent maintenance products where IT-systems are integral parts, for the purpose of securing availability and the intended use in the object business." (Nordström & Welanders, 2005, s. 4).

Motivation:

I selected this quote because it emphasizes the business-oriented view of system maintenance. Unlike the more technical definitions, Nordström & Welanders stress that IT systems must be maintained in line with business needs, balancing both stability and change.

Quote 3 – Riggs (1969)

“Systems maintenance, as I use the term, is the activity associated with keeping operational computer systems continuously in tune with the requirements of users, data processing operations, associated clerical functions, and external demands from governmental and other agencies.” (Riggs, 1969, s. 227).

Motivation:

I chose this quote because it provides a clear, practical definition of system maintenance from the perspective of daily IT operations. Riggs emphasizes that maintenance is not just technical repairs, but also ensuring systems evolve alongside user and regulatory demands.

IT Service Management (ITSM)

Quote 1 – Knapp (2010)

“IT service management (ITSM) is an integrated process approach that enables an IT organization to deliver services that meet business and customer requirements” (Knapp, 2010, s. 1).

Motivation:

I chose this quote because it provides a precise and widely recognized definition of ITSM. It highlights that ITSM is not just about isolated tasks but an integrated process approach. By stressing the alignment with both business and customer requirements, this definition clearly shows the strategic role of ITSM in organizations, which goes beyond technical operations.

Quote 2 – Thejendra (2014)

“The ITIL definition of IT service management is: ‘The implementation and management of quality IT services that meet the needs of the business. IT service management is performed by IT service providers through an appropriate mix of people, process and information technology.’” (Thejendra, 2014, s. 7).

Motivation:

This is a practical and business-aligned definition. It underlines ITSM’s core purpose, ensuring IT services are reliable, efficient, and tailored to business requirements. It also stresses the balance between people, process, and technology.

Reference list

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